Technical Response to

Decidability of $\mathcal{ALCI}_{RCC5}$ via Two-Tier Quotient Construction

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Abstract

This note records my assessment of the manuscript *Decidability of $\mathcal{ALCI}_{RCC5}$ via Two-Tier Quotient Construction*. The paper contains a promising and genuinely interesting idea: isolate the eventually periodic behavior of infinite PP-chains by finite period descriptors, and then try to manage cross-chain interaction at a separate quotient level. I do not, however, think that the manuscript establishes its main theorem. In my view, five technical issues remain open: (i) the kernel type $Core(Desc)$ forgets phase-specific obligations that later reappear in the soundness proof; (ii) the extension condition $V6$ filters only by RCC5 composition and not by description-logic type safety; (iii) the quotient uses reflexive PP-kernel loops under strong-EQ semantics without a proved translation back to a genuine RCC5 network; (iv) the finiteness argument for regular nodes still relies on an unproved redirection principle; and (v) the manuscript does not show that one period descriptor determines stabilized external behavior. For these reasons, I regard the two-tier quotient as promising partial progress, but not yet as a completed decidability proof.

1. Overall assessment

The manuscript addresses the precise gap that Wessel left open for $\mathcal{ALCI}_{RCC5}$: the logic has no finite model property, standard tree-style arguments fail, and a successful proof must reconcile complete-graph semantics with possibly infinite PP/PPI behavior. Wessel's report explicitly lists the decidability status of $\mathcal{ALCI}_{RCC5}$ as unknown and highlights the need for stronger machinery than the then available tableaux and lower-bound arguments [3].

Against that background, the present manuscript is ambitious and worthwhile. Its central idea — a two-tier quotient in which infinite PP-tails are compressed into finite descriptors, while cross-tail interaction is handled separately — is materially better than a naive replay or plain blocking argument. I also find the use of full RCC5 tractability conceptually natural: if one could reduce the quotient-realization step to a correctly specified path-consistent disjunctive RCC5 network, then the Renz–Nebel theorem would indeed be the right patchwork principle to invoke.

My disagreement is therefore not with the overall direction, but with the claim that the current manuscript already proves Theorem 9.1. In its present form, I believe the paper contains serious proof gaps, some in the extraction direction and some in the realization direction.

2. What is genuinely valuable in the paper

Before turning to the objections, it is important to record what I think the paper gets right.

1. The decomposition into *within-chain* periodic structure and *cross-chain* interaction is a real conceptual improvement over earlier one-level quotient proposals.
2. The emphasis on eventually recurrent PP-types and stabilized external behavior is exactly where any serious proof for $\mathcal{ALCI}_{RCC5}$ has to work.
3. The manuscript is refreshingly explicit about the role of full RCC5 tractability. If a sound quotient can be defined, path-consistency is indeed the right local criterion to aim for [2].

These points make the paper worth preserving and revising. The objections below are therefore not meant to dismiss the program, but to identify what still has to be repaired before a decidability theorem can be claimed.

3. Main technical concerns

3.1 The kernel core loses phase-specific obligations

In Definition 6.2, each period descriptor $Desc_\alpha = (\tau_1, \ldots, \tau_p)$ is assigned a single kernel node k_α with

$$tp(k_\alpha) = Core(Desc_\alpha) = \bigcap_j \tau_j.$$

Designated non-PP witnesses are then required only for existential demands occurring in this core. Later, in Theorem 8.1, Step 5, non-PP demands of chain elements are handled exactly through those designated witnesses.

That is not enough. A periodic tail may cycle through phase types $\tau_A, \tau_B, \ldots$ where some obligation occurs only in one phase. For example, one may have

$$\exists DR.D \in \tau_A \setminus \tau_B, \qquad \text{or} \qquad \forall PO.E \in \tau_A \setminus \tau_B.$$

Such formulas are perfectly compatible with the manuscript's descriptor conditions $(V1)$–$(V4)$ in Section 5, because those conditions constrain only the PP/PPI behavior internal to the descriptor. But once the quotient replaces the tail by a single kernel of type $Core(Desc_\alpha)$, the phase-specific obligations disappear from the data carried by the quotient.

This matters in the soundness direction. Theorem 8.1 unfolds a descriptor into an infinite chain whose positions are labelled by the original τ_i, but the quotient only guarantees designated non-PP witnesses for formulas in the *core*. Therefore the proof no longer establishes that a position of type τ_A satisfies a formula such as $\exists DR.D \in \tau_A \setminus Core(Desc_\alpha)$, nor does it ensure that a phase-specific universal such as $\forall PO.E$ is respected.

In short: the quotient is currently too coarse. Either the descriptor must retain phase-indexed obligations and phase-indexed designated witnesses, or the paper must prove that every non-PP demand needed in the realization migrates into the core. The manuscript does neither.

3.2 The extension condition $V6$ is RCC-consistent but not DL-safe

Definition 6.2 initializes the off-chain extension domains by

$$D(w, e) = comp(\text{demanded}, \rho(\text{parent}, e)),$$

and then requires path-consistency after arc-consistency enforcement. This is a purely RCC5-side filter. It checks composition compatibility, but it does *not* check whether the chosen relation is compatible with the concept labels carried by the endpoints.

A simple schematic example shows the problem. Suppose an existing node e carries

$$\forall PO.\neg A,$$

and the new witness w has

$$A \in tp(w).$$

Then $PO(w, e)$ is forbidden by the description-logic semantics: if w and e are PO-related, the universal on e forces $\neg A$ at w. However, the initialization rule for $D(w, e)$ will still admit PO whenever $PO \in comp(\text{demanded}, \rho(\text{parent}, e))$. Path-consistency alone will not remove that value unless some further RCC5 triangle blocks it. Consequently, a path-consistent atomic refinement may still violate the types.

This means that Theorem 8.1, Steps 4 and 5, do not yet show what they claim. The quotient may have an RCC-consistent extension that is not a DL-model. To repair this, the domains in $V6$ must be intersected with a *type-safety filter*. Concretely, for every pair of endpoint types (τ_w, τ_e) one needs the set of all relations R such that the universal formulas in τ_w and τ_e are simultaneously respected if R is chosen. Only then would path-consistency be relevant for the right network.

3.3 Reflexive PP-kernel loops are not legal strong-EQ diagonals

The quotient introduces one kernel node per descriptor and stipulates

$$\rho(k_\alpha, k_\alpha) = PP.$$

Section 3 justifies this by appealing to an algebraic self-absorption property of PP. But under the paper's own strong-EQ semantics, genuine model diagonals satisfy

$$\rho(d, d) = EQ,$$

not PP. A reflexive PP-loop may be a useful *abstract bookkeeping device*, but it is not itself a legal RCC5 edge in the target semantics.

This would be harmless if the paper gave a translation theorem from such abstract quotient gadgets to an ordinary RCC5 network with genuine EQ diagonals. But that bridge is missing. Instead, the quotient is subsequently treated as though it were already a suitable base network for path-consistency and realization. In particular, the paper invokes the full RCC5 tractability theorem on structures whose kernel self-loops are not obviously legal RCC5 constraints in the intended semantics.

So the issue is not merely cosmetic. The manuscript needs an explicit proof that every valid PP-kernel quotient can be transformed into a genuine strong-EQ RCC5 network to which Theorem 2.1 really applies.

3.4 The finiteness proof for regular nodes still hides an unproved blocking theorem

Theorem 7.1, Step 4, bounds the number of regular nodes by identifying two off-chain witnesses whenever they have

(i) the same Hintikka type,

(ii) the same demanded relation to their parent, and

(iii) the same stabilized relations to all kernel nodes.

The paper then states that once such a class repeats on a branch, the subtree below the repeat can be redirected to the earlier occurrence.

But in complete-graph semantics, that is precisely the hard point. Relations to the kernel nodes are not the whole context. Off-chain witnesses are also constrained by their relations to earlier regular nodes, siblings, and nodes created elsewhere in the already built part of the quotient. The blocking key used in Step 4 forgets those constraints, yet the proof assumes they can be ignored when redirecting a subtree.

This is the same structural obstacle that appears in every attempted blocking proof for $\mathcal{ALCI}_{RCC5}$:

the successor of a node is constrained by the *entire older complete graph*, not merely by one predecessor or by the kernel part. Unless the paper proves that kernel relations already determine all relevant regular-to-regular interaction, the redirection principle is not established.

So the claimed computable bound on regular nodes remains conditional on a missing theorem.

3.5 A descriptor alone does not yet determine stabilized external behavior

Theorem 7.1, Step 2, creates one kernel node per distinct period descriptor. For two kernels k_α, k_β, the paper defines $\rho(k_\alpha, k_\beta)$ as the doubly stabilized relation between two represented chains. This presupposes that a descriptor determines the stabilized external interface of its chain.

At present, that is not proved. Two chains may share the same eventually recurrent sequence of local types while still differing in how they interact with other chains or with regular nodes. The manuscript never establishes a uniqueness theorem saying that once the descriptor is fixed, all stabilized external relations are forced. Without such a result, one kernel per descriptor is too coarse: the quotient may identify chains whose external behavior is not interchangeable.

This concern is closely connected to the previous one. If external behavior is not descriptor-determined, then neither the kernel relation table nor the regular-node blocking classes are fine enough for a sound quotient.

4. Consequence for the main theorem

The paper's main result, Theorem 9.1, depends on two substantive directions:

1. **Extraction:** every genuine model yields a finite valid quotient (Theorem 7.1).
2. **Realization:** every finite valid quotient unfolds into a genuine model (Theorem 8.1).

In my reading, the objections above affect both directions. The kernel-core problem and the descriptor-interface problem undermine the quotient's adequacy as an extracted invariant. The regular-node finiteness argument still assumes an unproved blocking/redirection step. On the realization side, $V6$ currently enforces only RCC5 consistency, not type safety, and the quotient uses PP-kernel loops without a proved translation back to a legal strong-EQ network.

For that reason, I do not think Theorem 9.1 is established. The paper contains an interesting program for a proof, but not yet a complete proof.

5. What would make the approach convincing

I would see a realistic path to a revised paper if the following points were addressed explicitly.

1. **Retain phase-indexed obligations.** Period descriptors should remember not only the intersection $Core(Desc)$, but also enough phase information to replay all existential and universal requirements that actually occur on the tail.
2. **Make $V6$ DL-aware.** Initial domains for new witnesses should be filtered simultaneously by RCC5 composition and by the universal restrictions carried by endpoint types.
3. **Translate PP-kernels to legal RCC5 networks.** If reflexive PP is only an abstract quotient device, the paper should prove a theorem replacing it by a genuine strong-EQ representation before invoking RCC5 tractability.
4. **Prove the redirection theorem.** The finite regular-node bound needs a fully explicit theorem showing that the blocking class used in Step 4 preserves all constraints relevant to subtree reuse.
5. **Strengthen the interface invariant.** Either prove that a descriptor determines stabilized external behavior, or refine the quotient so that kernels are indexed by a richer interface type than the period descriptor alone.

If those issues were resolved, I would regard the two-tier quotient strategy as genuinely plausible. At present, however, the manuscript should, in my opinion, be described as promising progress rather than as a completed decidability proof.

6. Conclusion

My overall judgment is therefore the following. The manuscript contains a serious and potentially important idea, and it improves substantially on simpler quotient and replay proposals. But the current proof of decidability is not yet complete. The right headline is, in my view, not "the decidability of $\mathcal{ALCI}_{RCC5}$ is now proved," but rather "a promising quotient architecture has been identified, and the remaining proof obligations are now much sharper than before."

That is already worthwhile progress. It simply should not be overstated.

References

[1] Claude. *Decidability of $\mathcal{ALCI}_{RCC5}$ via Two-Tier Quotient Construction: Period descriptors, PP-kernel nodes, and the full tractability of RCC5.* Discussion draft, April 2026.

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[3] Michael Wessel. *Qualitative Spatial Reasoning with the ALCIRCC Family – First Results and Unanswered Questions.* Technical Report FBI-HH-M-324/03, University of Hamburg, 2002/2003.